**D 10671** **(Pages : 3)** **Name.....**

**Reg. No.....**

**FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021**

(CBCSS—UG)

Mathematics

MTS 5D 01—APPLIED CALCULUS

(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

**Section A**

*Answer at least* **eight** *questions.*
*Each question carries* 3 *marks.*
*All questions can be attended.*
*Overall Ceiling* 24.

1. If $g(t)=(t-2)^{\frac{1}{2}}$, find $g(27)$ and $g(5)$?

2. Find the domain and range of $f(x)=\frac{1}{x-3}$.

3. Find $f(g(x))$ where $f(u)=u^2+3u+1$ and $g(x)=x+1$.

4. Find the distance between the points $P(-2,5)$ and $Q(4,-1)$.

5. State vertical line test.

6. What is slope of (a) X-axis ; and (b) Y- axis.

7. Find the equation of the line that passes through the points $(3,-2)$ and $(1,6)$.

8. Find $\lim_{x\to 1}\frac{x^3-8}{x-2}$.

9. Differentiate the polynomial $y=5x^3-4x^2+12x-8$.

**Turn over**

10. Find all critical numbers of the function $f(x) = 2x^4 - 4x^2 + 3$.

11. Find all real numbers $x$ that satisfy the given equation $4^{2x-1} = 16$.

12. Evaluate $\log_2 32$.

(8 × 3 = 24 marks)

## Section B

*Answer at least* **five** *questions.*
*Each question carries* 5 *marks.*
*All questions can be attended.*
*Overall Ceiling* 25.

13. Find :

(a) $\lim_{x \to \infty} \frac{2x^2 + 3x + 1}{3x^2 - 5x + 2}$.

(b) $\lim_{x \to 1} \frac{x^2 - 1}{x^2 - 3x + 2}$.

14. For the function $f(x) = 1 - x^2, 0 \le x < 2$
$= 2x + 1, x \ge 2.$

evaluate the one sided limits $\lim_{x \to 2-} f(x)$ and $\lim_{x \to 2+} f(x)$.

15. Find the equation of the tangent line to the curve $y = \sqrt{x}$ at the point where $x = 4$.

16. Find the intervals of increase and decrease for the function $x^2 - 4x + 5$.

17. If $f(x) = 5^{x^2 + 2x}$ find all values of $x$ such that $f(x) = 125$.

18. Discuss the continuity of the function $f(x)=\frac{x+2}{x-3}$ on $-2<x<3$ and on the closed interval $-2\le x\le 3$.

19. The Gross Domestic Product (GDP) of a certain country was $N(t)=t^2+5t+106$ billion dollars $t$ years after 1998 :

    (a) At what rate was the GDP changing with respect to time in 2008.

    (b) At what percentage rate was the GDP changing with respect to time in 2008.

(5 × 5 = 25 marks)

## Section C

*Answer any* **one** *question.*
*The question carries* 11 *marks.*

20. Graph the function $f(x)=-x^2+x+2$. Include all $x$ and $y$ intercepts.

21. A manufacturer's total cost consists of a fixed overhead of \$200 plus production cost of \$50 per unit. Explore the total cost as a function of the number of units produced and draw graph.

(1 × 11 = 11 marks)